Additive Manufacturing of Toroid Inductor for Power Electronics Applications

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***Abstract*—To increase power density and simplify the process of integrating magnetic components to power electronics circuits, a commercial multiextruder paste-extrusion three-dimensional (3-D) printer was used to process both metal and magnetic pastes into 3-D structures of magnetic components. A magnetic-filled-benzocyclobutene composite, termed poly-mag paste, was formulated and used as a magnetic core feedstock for the printer, and a commercial nanosilver paste was used as a conductive winding feedstock for the printer. A toroid inductor was 3-D printed by using metal and magnetic pastes, and it was cured at 250 °C for a half hour without any external pressure to form the structure. The inductance of the 3-D-printed toroid inductor was measured to be 167–143 nH in the frequency range of 1 kHz to 3 MHz. The dc resistance of the winding was 0.1 Ω. These results are in good agreement with a finite-element analysis simulation. Both the winding and core magnetic properties can be improved by adjusting the feed paste's formulations and flow characteristics, and by fine tuning the printer parameters, such as motor speeds, extrusion rate, and nozzle sizes.**

***Index Terms*—Additive manufacturing (AM), magnetic components, magnetic-filled polymer composite paste, nanosilver paste, power electronics integration.**

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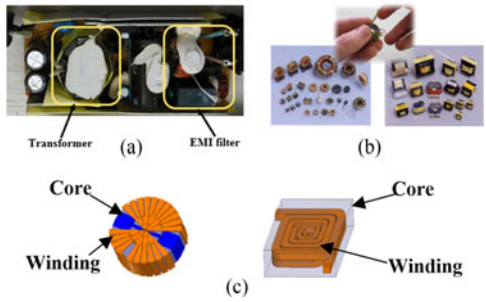


Fig. 1. (a) Typical power supply consisting of a number of discrete magnetic components. (b) Basic process for making the windings of inductors and transformers. (c) Integrated uniform-flux inductor design.

I. INTRODUCTION

HETEROGENEOUS integration of magnetic components, such as inductors and transformers, as shown in Fig. 1(a), has the potential to further increase power density and efficiency in power electronic converters [1]. However, today's magnetic cores and electrical conductor windings that make up magnetic components are fabricated separately in multiple complex steps. For example, to make a soft ferrite inductor, the core is first formed by a high-pressure powder-compaction process followed by sintering at high temperatures (typically >900 °C), and then copper wires are "hand wound" around the core, as shown in Fig. 1(b) [2]–[4]. This manufacturing approach gives rise to bulky discrete components and precludes the implementation of high-density high-performance integrated geometries. For example, the constant-flux magnetic component structures shown in Fig. 1(c) cannot be fabricated by current manufacturing processes [5], [6]. Thus, to improve power conversion efficiency, power density, and reliability of a power electronics system, there is a need for manufacturing technologies that would enable ease of integration of the magnetic components.

One possible way to integrate the inductor/transformer into modules is to use an additive manufacturing (AM) process to fabricate the magnetics. Three-dimensional printing is a layer-by-layer process of making products and components from a digital model. Its potential has been demonstrated for applications in various industries. Some key benefits of AM are shorter lead times, mass customization, reduced parts count, more complex shapes, less material waste, and lower life-cycle energy use [7]. Recently, some research groups explored application of AM in power electronics. For example, Oak Ridge National

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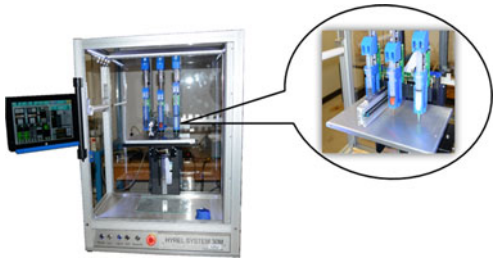


Fig. 2. Multiextruder paste-extrusion 3-D printer.

Laboratory (ORNL) researchers [8], [9] reported in 2014 a 3-D-printed aluminum heat sink and a plastic lead frame for a 10-kW power inverter, and then in 2016, a ferrite E-core with relative permeability of about 2; Liang *et al*. [10] 3-D printed a plastic structure as a mold for making a unique sterling-silver inductor; Proto-pasta [11] introduced an iron-powder-filled polylactic-acid filament for use with a fused-deposition-model (FDM) 3-D printer to make magnetic cores with relative permeability of about 1.5; and Wang *et al.* [12] developed a NiZn-ferrite-filled ABS filament for the FDM printer with relative permeability of about 2.

Magnetic components in most power converters have complex configurations with irregular core shapes and intricate winding structures. Although 3-D printing is ideal for making complex and intricate structures, fabricating parts containing different types of materials remain to be a challenge for AM. We chose to work with a commercial multimaterial extrusion-based 3-D printer because it offers ease and flexibility of coprocessing multiple materials. We have formulated feedstock pastes to make them compatible with the 3-D printer. Use of paste as the feedstock reduces material waste, lowers the equipment cost, and simplifies the part-construction process. The paste-based AM process can be readily scaled up to manufacture a multimaterial and multifunctional system. Thus, this 3-D-printing platform offers the potential for further integration of a power electronics circuit by concurrent manufacturing of capacitive, magnetic, and resistive components. In 2016, our research group was the first to report [13] the feasibility of fabricating magnetic components with both winding and magnetic cores by using the 3-D printer. The feed-materials for the printer were a custom-formulated low-temperature (<250 °C) curable magnetic paste and a commercial nanosilver paste invented in our laboratory. The relative permeability of the low-temperature curable magnetic paste was about 10, and printing of a planar inductor with simple structure was demonstrated. In this study, we reformulated the magnetic paste to increase its permeability and improved the printing parameters to fabricate the more complex-shaped toroid inductor [14].

II. Equipment and Materials for Additive Manufacturing of Magnetic Components

A. Commercial Multiextruder Paste-Extrusion 3-D Printer

The multiextruder paste-extrusion 3-D printer shown in Fig. 2 was purchased from Hyrel 3D [15]. It has a feed-cartridge assembly with four extruders (only three are shown in the figure) that can print four different paste materials stored in syringes.

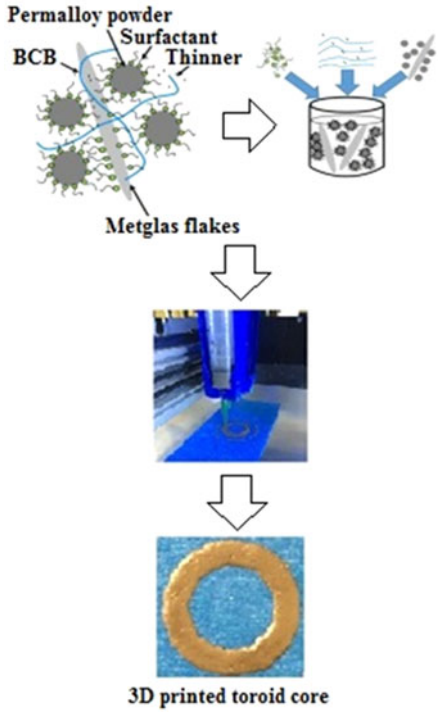


Fig. 3. Low-temperature curable poly-mag paste for paste-extrusion 3-D printer.

B. Poly-Mag Paste

A low-temperature (<250 °C) curable magnetic paste consisting of soft magnetic fillers and a thermoset polymer called poly-mag paste was formulated by extending our prior research experience in the development of the nanosilver paste. The magnetic fillers used in the paste formulation were round-shaped particles of permalloy and flake-shaped particles of glassy alloy. The permalloy particles (Permalloy-80) with an average size of 12 μm was purchased from ESPI Metals. The glassy-alloy particles were obtained by grounding 2705M ribbon (Metglas, Inc.) with a 6770 SPEX Sample prep freezer/mill cryogenic grinder. The milled flakes have a length and width ranging from 75 to 250 μm and the thickness of around 25 μm. The thermoset polymer is Cyclotene 3022, Benzocyclobutene (BCB) obtained from Dow Chemical Company. BCB is widely used in the electronics industry due to its ease of processing, low dielectric constant (2.7), and good thermal stability [16]. The Poly-mag paste consisted of 10 weight percent of BCB and 90 weight percent of magnetic fillers. BCB was added as a binder to ensure close packing of magnetic fillers and sufficient mechanical strength for handling. The remaining magnetic fillers consisted of a mixture of 90 weight percent permalloy particles and 10 weight percent Metglas flakes. Other organic components, such as dispersant and solvent to create uniform suspensions of the magnetic fillers in the organic system, were added to produce the paste with good particle dispersion, flowability, and viscosity. Fig. 3 illustrates some major ingredients in the poly-mag paste,

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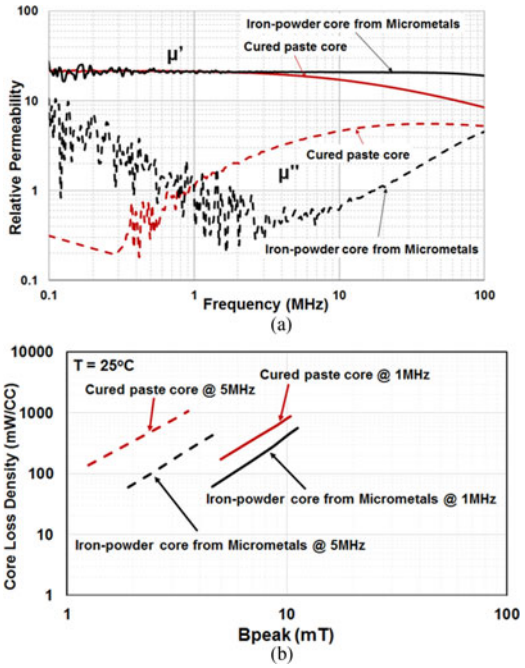


Fig. 4. (a) Complex permeability dispersion spectra of our magnetic paste toroid core and the iron-powder toroid core from Micrometals. (b) Core-loss density of our magnetic paste toroid core and the Micrometals core at frequency of 1 and 5 MHz.

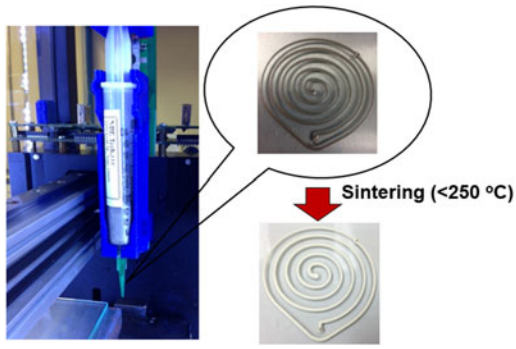


Fig. 5. Commercial nanosilver paste for paste-extrusion 3-D printer.

TABLE I
ELECTRICAL RESISTIVITY OF LOW-TEMPERATURE SINTERED 3-D-PRINTED WINDING AND THAT OF BULK SILVER

Materials	Electrical resistivity (10^{-8} Ω·m)
250 °C sintered 3-D-printed silver winding	4.8
Pure silver wire	1.6

which can be extruded out to form magnetic core structure by the 3-D printer.

To investigate the magnetic properties of the poly-mag paste, the paste was fabricated into toroid core shapes by an injection molding process. The magnetic paste was squeezed into a toroid-core-shaped silicone rubber mold and then formed by a vacuum-assisted process. The toroid core shape paste was cured at 250 °C for 30 min at a heating rate of 1 °C/min in programmable muffle furnace. The as-fabricated core had the following dimensions: outer diameter = 20 mm; inner diameter = 14.9 mm; height = 6 mm. For comparison, we obtained a custom-made iron-powder core from Micrometals (Anaheim, CA, USA), which had the same dimensions as our cured toroid core but was manufactured under high external pressure and high temperature.

Measurements of complex permeability versus frequency from 100 kHz to 110 MHz at zero dc bias on the cured paste core and the Micrometals core were done by using a precision impedance analyzer (4294A; Agilent, Santa Clara, CA, USA) with a magnetic material test fixture (16454A; Agilent, Santa Clara, CA, USA). The core-loss density measurements of these two cores were taken at 1 and 5 MHz by using a setup described in [17] and [18].

Fig. 4(a) shows plots of complex permeability versus frequency obtained on the cured paste core and the Micrometals core. Both cores had the same real parts (μ') of permeability of 23, but the cured paste core with Metglas addition had a narrower bandwidth and much higher μ'' than those of the Micrometals core.

The μ'' of the complex permeability of a magnetic component measured by an impedance analyzer is generally attributed to energy loss mechanisms in the magnetic material, but the alternating magnetic field excitation was limited by the power capability of the impedance analyzer. The core-loss density or power loss per-unit volume versus alternating B-field strength at a given frequency is a more relevant property of a magnetic component when considering its use in a converter. Fig. 4(b) shows the core-loss density plots measured at 1 and 5 MHz of the two cores at room temperature. The figure shows that of each core, the core-loss density at 5 MHz is nearly ten times higher than that at 1 MHz; and the core-loss density of the cured paste core is about three times higher than that of the Micrometals core.

C. Nanosilver Paste

To print electrical winding, we tested the commercial nanosilver paste which was developed for chip bonding [19]. The material can be sintered at temperatures below 250 °C to result in excellent thermal and electrical properties [20]. Fig. 5 shows that the nanosilver paste can be adapted to the 3-D printing platform and extruded out to form a winding structure. The nanosilver paste can achieve a crack-free 3-D-printed single-layer thicker than 200 μm. Table I lists the electrical resistivity of the 3-D-printed winding after low-temperature sintering process. Currently, the electrical resistivity of the low-temperature sintered 3-D-printed winding is about three times higher than of the pure silver wire.

We recognize the need for drastically reducing the winding resistivity if the 3-D-printed magnetic components are of any practical use. Researchers in AM are actively pursuing ways to increase electrical conductivities of 3-D-printed metals. One possible way is to use laser-assisted sintering. In this technique, a laser system is added to a 3-D printer [21], [22]. By directing a focused laser beam on the metal as it is being printed, the material can be heated locally to a high temperature to speed up the consolidation of the metal particles resulting in higher electrical conductivity. The local heating prevents possible adverse impact on the printed magnetic material. Another way to improve the electrical conductivity is by microwave-assisted sintering

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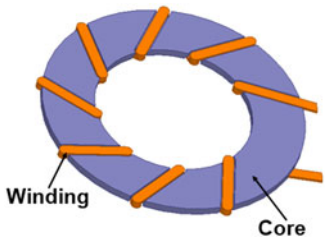


Fig. 6. Design of toroid inductor for 3-D printing.

[23]. In this way, the entire printed parts are heated in an oven using both conventional heat and a microwave. By adjusting the microwave frequency to maximize its absorption by the metal, the metal winding can be heated over a short period of time to a much higher temperature than the magnetic material. We have plans to explore both of these techniques to increase the winding conductivity.

III. Structure Design and Feasibility Demonstration of 3-D Printing a Toroid Inductor

A. Structure Design

At present, almost all of the toroid inductors used in PE circuits are fabricated in a way that involves making the cores first by metal casting and/or powder processing and then combining with a winding of metal traces or wires. To demonstrate the feasibility of fabricating complex magnetic component structures with the Hyrel 3D system, we tested it by printing a toroid inductor. Fig. 6 shows the design of a toroid inductor.

The inductor has eight turns of winding and part of its winding buried inside the core. The buried winding can serves as supporting material during the printing process. The designed toroid inductor had the following dimensions: 1) outer diameter of the core = 20 mm; 2) inner diameter of the core = 12 mm; 3) thickness of the core = 0.6 mm; 4) winding width = 0.8 mm; and 5) winding thickness = 0.2 mm.

B. Fabrication Procedures

Fig. 7 shows the process flow for making a 3-D-printed toroid core: (a) Insertion of the two paste feedstock, the poly-mag and nanosilver paste; (b) printing of a base layer of the poly-mag paste with gaps left for winding; (c) the base layer with the silver winding printed to fill the gaps; (d) printing of two more layers of the poly-mag paste to achieve a designed thickness; (e) final deposition of the nanosilver paste to connect the winding on the base layer to the top. The thickness of each layer poly-mag or nanosilver paste was 0.2 mm. During the entire printing process, the build-plate temperature was set at 70 °C. After the core was printed, it was transferred to a programmable muffle furnace to simultaneously cure the polymer in the magnetic core and sinter the nanosilver winding. The curing profile is shown in Fig. 7(f). Fig. 7(g) shows the finished 3-D-printed toroid inductor after polishing.

The printed toroid inductor had the following dimensions: outer diameter of the core = 19.8 mm; inner diameter of the core = 12.2 mm; thickness of the core = 0.55 mm; average winding width = 0.88 mm; and average winding thickness = 0.18 mm. The printed dimensions are within 10% error than those of the designed structure.

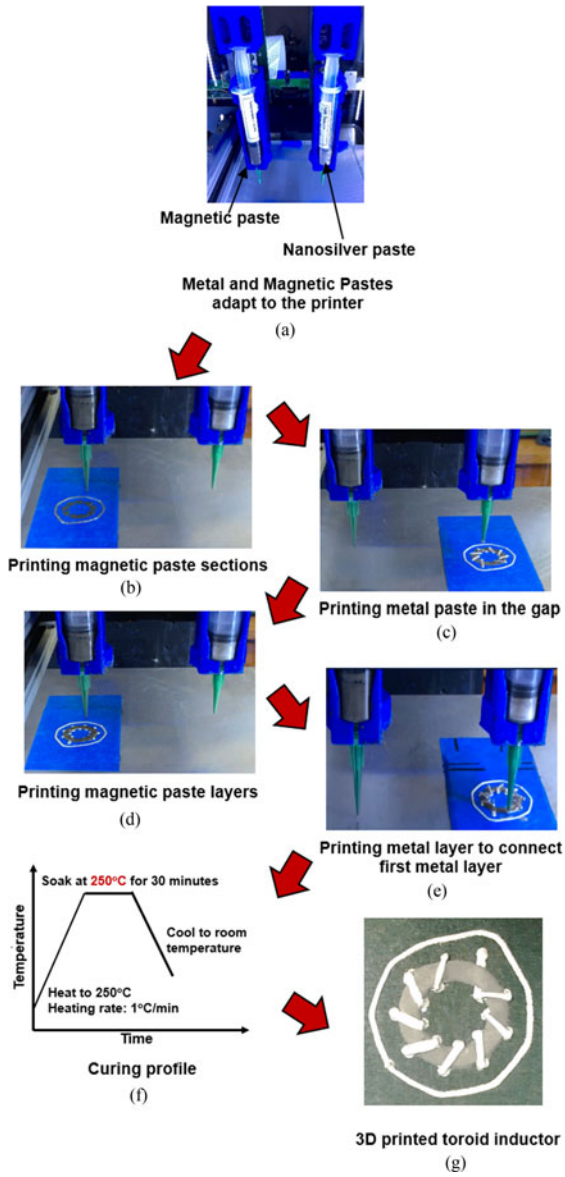


Fig. 7. Process of 3-D printing toroid inductor: (a) metal and magnetic pastes adapt to the printer; (b) printing magnetic paste sections; (c) printing metal paste in the gap; (d) printing magnetic paste layers; (e) printing metal layer to connect first metal layer; (f) curing profile; and (g) 3-D printed toroid inductor.

IV. Characterization and Discussion

A. Inductance and DC Winding Resistance of the 3-D-Printed Toroid Inductor

The layer-by-layer process of making the toroid core requires burying a portion of the winding on the base layer inside the magnetic material. This complicates calculations of the inductance and dc winding resistance from simple equations. Therefore, we ran a finite-element analysis (FEA) to simulate the component properties. In the analysis, the winding electrical conductivity was set to be three times lower than that of bulk silver. The relative permeability of the magnetic material used in the simulation was 23. This value was based on the

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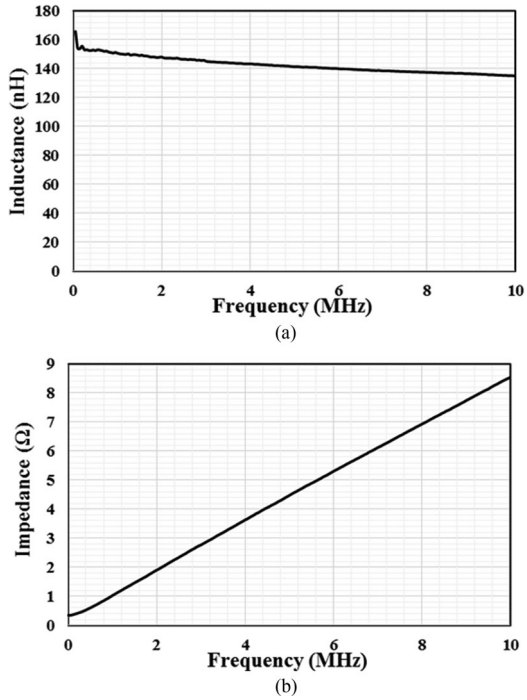


Fig. 8. (a) Inductance versus frequency at zero dc bias of the 3-D-printed toroid core. (b) Impedance versus frequency of the 3-D-printed toroid core.

measurements of the material's complex permeability in the frequency range of 1 kHz to 3 MHz. The simulated inductance and dc winding resistance of the inductor were 160 nH and 0.08 Ω, respectively.

A high-precision impedance analyzer was used to characterize the properties of the 3-D-printed toroid inductor. Fig. 8(a) is a plot of the measured inductance versus frequency at zero dc bias, and Fig. 8(b) shows the impedance versus frequency of the inductor.

The measured inductance is 167 to 143 nH in the frequency range of 1 kHz to 3 MHz. The inductance at the lower frequency end is in good agreement with the simulated result. The measured inductance decreases with increasing frequency, which can be attributed to a decreasing relative permeability of the magnetic material at higher frequency. The measured dc winding resistance of the inductor is 0.1 Ω, which is slightly higher than the simulated. This is likely due to the contact resistance between the measurement probes and the printed silver.

B. Microstructure

It is well known that properties of a material are determined by its microstructure. Examination of the microstructure of the 3-D-printed toroid core can help gaining insights for ways to improve the magnetic properties. Fig. 9 is a scanning-electron-microscope (SEM) image of a 3-D-printed toroid core. One can see the magnetic fillers and voids. The voids, likely created by trapped air bubbles in the poly-mag paste, lowered the relative permeability of the core. The random distribution of the Metglas flakes is another reason for the low relative permeability. The Metglas flakes used in this study were made by grinding a Metglas ribbon of over 25 μm thick. The grinding process was only able to break the ribbon into a powder, but not reducing its

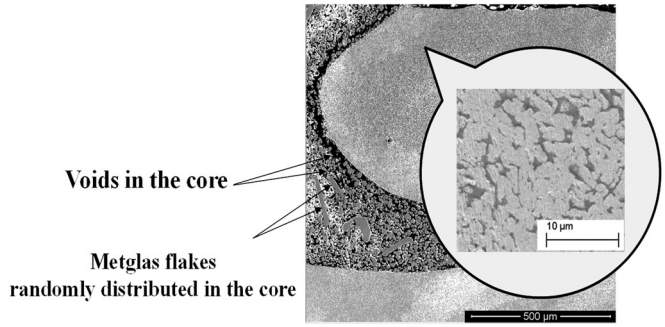


Fig. 9. SEM image of the 3-D-printed toroid inductor.

thickness. Because the Metglas material is highly conductive, we believe that the interflake eddy-current loss is a significant contributor to the higher core-loss density found in the cured poly-mag cores.

A few approaches can be taken to improve the performance of the printed inductor. For example, the curing profile can be optimized to reduce the void content in the magnetic core, and, thus, increase the relative permeability. Aligning the Metglas flakes along the magnetic flux direction can also increase relative permeability. To effectively reduce the core-loss density, we would need to use thinner Metglas flakes and aligning them along the flux direction [24]. To reduce the winding loss, as mentioned earlier, either laser- or microwave-assisted sintering technique can be explored to further densify the sintered microstructure of the printed metal.

V. Summary and Future Work

The feasibility of AM of complex magnetic component by printing both the magnetic core and conductive winding using a commercial multiextruder paste-extrusion 3-D printer was demonstrated. We formulated a BCB-matrix-permalloy-Metglas composite for the magnetic feedstock. Such composite has relative permeability of 23 after curing at 250 °C. The formulation makes it possible to simultaneously cure the magnetic paste and sinter a nanosilver paste, which was used as a feedstock for the winding. A toroid inductor was designed and 3-D printed. The measured inductance and dc winding resistance of the printed toroid core were in good agreement with those simulated by a FEA of the inductor model.

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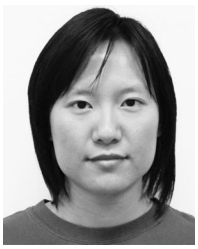
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